CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 22: Trigonometrical Ratios

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. In triangle ABC, $\angle B = 90^\circ$, $AB = 8$ cm, $BC = 6$ cm. Find $\sin A$ and $\cos A$.

2. If $\sin\theta = \frac{5}{13}$, find $\cos\theta$ and $\tan\theta$.

3. If $\tan A = \frac{3}{4}$, find $\sin A + \cos A$.

4. If $\sec\theta = \frac{25}{7}$, find $\tan\theta$ and $\operatorname{cosec}\theta$.

5. If $\cos\theta = \frac{3}{5}$, find $\frac{\sin\theta - \cot\theta}{2\tan\theta}$.

6. If $\tan\theta = \frac{5}{12}$, find $\frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta}$.

7. If $3\tan\theta = 4$, find $\frac{4\cos\theta - \sin\theta}{2\cos\theta + \sin\theta}$.

8. In triangle PQR, $\angle Q = 90^\circ$, $PR = 26$ cm and $\cos P = \frac{5}{13}$. Find PQ and QR.

9. If $\sin\theta = \cos\theta$ for an acute angle, find $\tan\theta$.

10. Is $\sin\theta = \frac{4}{3}$ possible for an acute angle? Why?

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Answer Key

1. $AC = 10$ cm; $\sin A = \frac{3}{5}$, $\cos A = \frac{4}{5}$
2. $\frac{12}{13}$ and $\frac{5}{12}$
3. $\frac{7}{5}$
4. $\frac{24}{7}$ and $\frac{25}{24}$
5. $\frac{\frac{4}{5} - \frac{3}{4}}{\frac{8}{3}} = \frac{3}{160}$
6. $-\frac{17}{7}$
7. $\frac{4 - \frac{4}{3}}{2 + \frac{4}{3}} = \frac{4}{5}$
8. 10 cm and 24 cm
9. 1
10. No; the perpendicular cannot be longer than the hypotenuse

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